

Embed the enhanced physics-informed neural network model into the aircraft lithium battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

Executive Technology Brief

Domain: thermal

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'thermal runaway' concerning aircraft lithium battery installation. User-submitted engineering problem (session input): "Why do aircraft lithium-battery installations suffer thermal events? (connector outage honesty)". [nasa_ntrs] 1 records retrieved live (2019-2019). Illustrative records: 'Statistical Characterization of 18650-Format Lithium-Ion Cell Thermal Runaway Energy Distributions'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Mechanical abuse poses a critical safety risk to Li-ion batteries by inducing internal short circuits that initiate thermal runaway. Remarkably, voltage recovery frequently emerges during thermal runa'; 'The pursuit of higher energy density in lithium-ion batteries has made high-nickel (Ni) layered oxides leading cathode candidates for next-generation electric vehicles. However, their poor thermal sta'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Incorporation of Joule heating, reversible entropic heat, and radiative losses into a physics-informed neural network for real-time temperature prediction.

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Embed the enhanced physics-informed neural network model into the aircraft lithium battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

Governing Model: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT .

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected).

Engaged variables: k , dT/dx .

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.